EX NO: 10	CALCULATION OF PROCESSING GAIN AND GENERATION OF PN SEQUENCES FOR SPREAD SPECTRUM COMMUNICATION USING MATLAB SIMULATION
Date:	

```
clc;

clear;

close all;


%% PN Sequence Lengths

PN_length = [7 15 31 63 127 255 511 1023];


%% Processing Gain (Linear)

PG_linear = PN_length;


%% Processing Gain in dB

PG_dB = 10 * log10(PG_linear);


%% Display Table

T = table(PN_length', PG_dB', ...

    'VariableNames', {'PN_Length', 'Processing_Gain_dB'});


disp('Processing Gain Table');

disp(T);


%% Plot

figure;


stem(PN_length, PG_dB, 'filled');
```

grid on;

xlabel('PN Sequence Length');

ylabel('Processing Gain (dB)');

title('Spread Spectrum: Processing Gain vs PN Sequence Length');

Processing Gain Table

PN_Length	Processing_Gain_dB
7	8.451
15	11.761
31	14.914
63	17.993
127	21.038
255	24.065
511	27.084
1023	30.099